

VisionInspect AI

Manufacturing Defect Detection & Quality Inspection System

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Milestone-3

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1. Introduction

VisionInspect AI is an AI-powered manufacturing quality inspection system designed to detect product defects, assess their severity, and support quality-control decision making.

Milestone 3 focused on transforming the existing inspection pipeline into a more complete production-oriented system by introducing:

- Persistent database storage
- Inspection history
- Analytics and production reports
- Authentication and user management
- Supervisor-level monitoring
- Inspection escalation and rework workflows
- Integration of backend analytics with the frontend dashboard

The milestone therefore moved the project beyond individual image prediction toward a system capable of maintaining historical inspection records and supporting operational decisions.

Authorize



default



POST	/upload	Upload Image	▼
GET	/dataset	Load Dataset	▼
POST	/preprocess	Preprocess	▼
POST	/augment	Augment Dataset	▼
GET	/severity	Get Severity	▼
GET	/statistics	Statistics	▼
GET	/	Home	▼
GET	/status	Get Status	▼
POST	/predict	Predict	▼
POST	/batch-predict	Batch Predict	▼
GET	/analytics/trends	Get Analytics Trends	▼
GET	/analytics/risk-assessment	Get Risk Assessment	▼
GET	/reports/production	Get Production Report	▼
GET	/report/{inspection_id}	Get Report	▼
POST	/inspect	Inspect Image	▼

POST	/inspect	Inspect Image	▼
GET	/history	Get History	▼
GET	/reports	Get Reports	▼
GET	/analytics	Get Database Analytics	▼
GET	/batches	Get Batches	▼
POST	/supervisor/inspections/{inspection_id}/approve-rework	Approve Rework	▼
POST	/supervisor/inspections/{inspection_id}/escalate	Escalate Inspection	▼
GET	/supervisor/workflows	Get Supervisor Workflows	▼

Authentication



POST	/auth/signup	Signup	▼
POST	/auth/login	Login	▼
GET	/auth/profile	Profile	🔒 ▼
GET	/auth/users	Get All Users	🔒 ▼
GET	/auth/users/{user_id}	Get User	🔒 ▼
PUT	/auth/users/{user_id}	Update User	🔒 ▼
DELETE	/auth/users/{user_id}	Delete User	🔒 ▼
GET	/auth/admin	Admin Dashboard	🔒 ▼

2. Milestone 3 Objectives

The major objectives of Milestone 3 were:

Persistent Data Storage

Inspection results needed to be stored permanently instead of existing only during an individual API request.

Historical Inspection Tracking

The system needed to maintain records containing information such as:

- Image name
- Product category
- Detected defect
- Inspection result
- Confidence
- Anomaly score
- Severity score
- Severity level
- User
- Timestamp

Analytics

The system needed to calculate production-level metrics such as:

- Total inspections

- Pass rate
- Defect rate
- Severity distribution
- Category-level performance
- Risk levels

Authentication

A database-backed authentication system was introduced to support user registration, login, profiles, and administrative operations.

Supervisor Dashboard

A dedicated supervisor view was developed to provide:

- Production KPIs
- Production-line health
- Pass-rate trends
- Defect distribution
- Critical inspection queue
- Rework approval
- Inspection escalation
- CSV export

Operational Workflow

Critical inspections could be acted upon by a supervisor through:

Pending → Rework Approved

or

Pending → Escalated

This established the foundation for a real quality-control workflow.

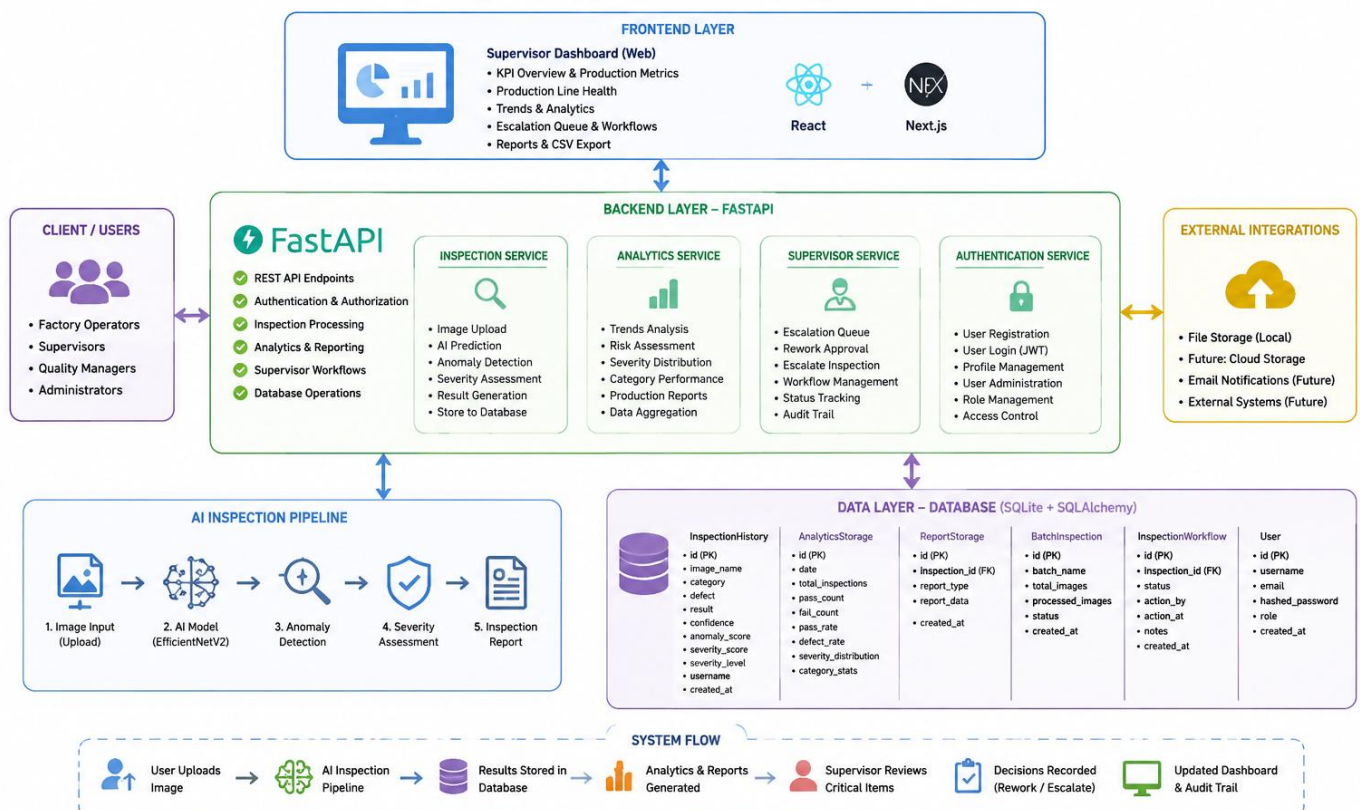
3. System Architecture

Milestone 3 uses a layered architecture consisting of the frontend, FastAPI backend, database layer, analytics services, and AI inspection pipeline.

Architecture Flow

VisionInspect AI – System Architecture (Milestone 3)

Database Integration, Analytics, Authentication & Supervisor Dashboard



The frontend communicates with FastAPI through REST APIs. FastAPI handles authentication, inspection operations, analytics, reports, and supervisor actions.

SQLAlchemy provides the database abstraction layer, while SQLite is used for local persistent storage.

4. Database Integration

A major part of Milestone 3 was integrating persistent database storage.

The project uses SQLite with SQLAlchemy.

The database configuration is provided through the environment file:

```
DATABASE_URL=sqlite:///./visioninspect.db
```

SQLAlchemy is used to define database models and manage database sessions.

The application initializes the database using:

```
Base.metadata.create_all(bind=engine)
```

This ensures that the required database tables are created when the backend starts.

Database Models

The milestone introduced and integrated several database models, including:

- InspectionHistory
- AnalyticsStorage
- ReportStorage
- BatchInspection
- InspectionWorkflow
- User

These models allow the system to store inspection results, analytics information, generated reports, batch records, workflow actions, and authentication information.

Inspection History Model

The InspectionHistory table stores the results of completed inspections.

Important fields include:

Field	Purpose
ID	Unique inspection identifier
Username	User who performed the inspection
Image Name	Name of inspected image
Category	Product category
Defect	Detected defect type
Result	PASS or REJECT

Field	Purpose
Confidence	Prediction confidence
Anomaly Score	Anomaly detection score
Severity Score	Calculated severity
Severity Level	Low/Medium/High/Critical
Created At	Inspection timestamp

This allows the system to retain complete historical inspection information.

5. Inspection History Management

The inspection pipeline was connected to the database so that inspection results are stored automatically.

The `/inspect` endpoint performs the following operations:

1. Receives the image.
2. Performs AI-based defect prediction.
3. Calculates the inspection result.
4. Calculates anomaly and severity information.
5. Stores the inspection in the database.
6. Generates the inspection report.
7. Returns the inspection result and report.

This creates a persistent connection between the AI inspection pipeline and the production database.

History API

The endpoint:

GET /history

retrieves previously stored inspection records.

```
category      : bottle
id            : 7
result        : PASS
anomaly_score : 0.3392
severity_level : Low
username      : admin1
image_name    : 003.png
defect        : good
confidence    : 99.0
severity_score : 5.64
created_at    : 2026-08-07T15:16:33.436488

category      : bottle
id            : 6
result        : PASS
anomaly_score : 0.33811
severity_level : Low
username      : admin1
image_name    : 005.png
defect        : good
confidence    : 99.0
severity_score : 5.61
created_at    : 2026-08-07T15:16:25.302542

category      : metalnut
id            : 5
result        : REJECT
anomaly_score : 0.39507
severity_level : Critical
username      : admin1
image_name    : 000.png
defect        : large_crack
confidence    : 99.0
severity_score : 97.16
created_at    : 2026-08-07T14:47:34.267403
```

6. Analytics and Reporting

Milestone 3 introduced database-driven analytics and reporting for production monitoring.

The system calculates:

- Total inspections
- Pass and defect rates
- Severity distribution
- Category-wise performance
- Inspection trends
- Production risk levels

The `/analytics/trends` endpoint provides inspection trends, while `/analytics/risk-assessment` classifies product categories as *LOW*, *MEDIUM* or *HIGH RISK* based on their defect rates.

The `/reports/production` endpoint generates a production quality summary containing total units inspected, passed/rejected units, pass rate, defect rate and severity distribution.

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/reports/production' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8080/reports/production

Server response

Code	Details
200	Response body <pre>{ "report_title": "Executive Production Quality Summary Report", "system_status": "OPERATIONAL", "total_units_inspected": 8, "units_passed": 7, "units_rejected": 1, "yield_pass_rate_pct": 87.5, "defect_rate_pct": 12.5, "severity_distribution": { "critical": 1, "High": 0, "Medium": 0, "Low": 7 }, "category_performance": { "bottle": { "total": 7, "anomalous": 0 }, "metalnut": { "total": 1, "anomalous": 1 } } }</pre> Response headers <pre>content-length: 368 content-type: application/json date: Sat, 08 Aug 2026 10:21:00 GMT server: uvicorn</pre>

Responses

Code	Description
200	Successful Response

7. Authentication and User Management

Database-backed authentication was integrated into the system during Milestone 3.

The authentication module supports:

- User signup and login
- JWT-based authentication
- User profiles
- User management
- Administrator access

Important endpoints include /auth/signup, /auth/login, /auth/profile, /auth/users and /auth/admin.

User information is stored in the database, providing persistent authentication and a foundation for future role-based access control.

8. Supervisor Dashboard

The Supervisor Dashboard provides a centralized view of production quality and critical inspection activity.

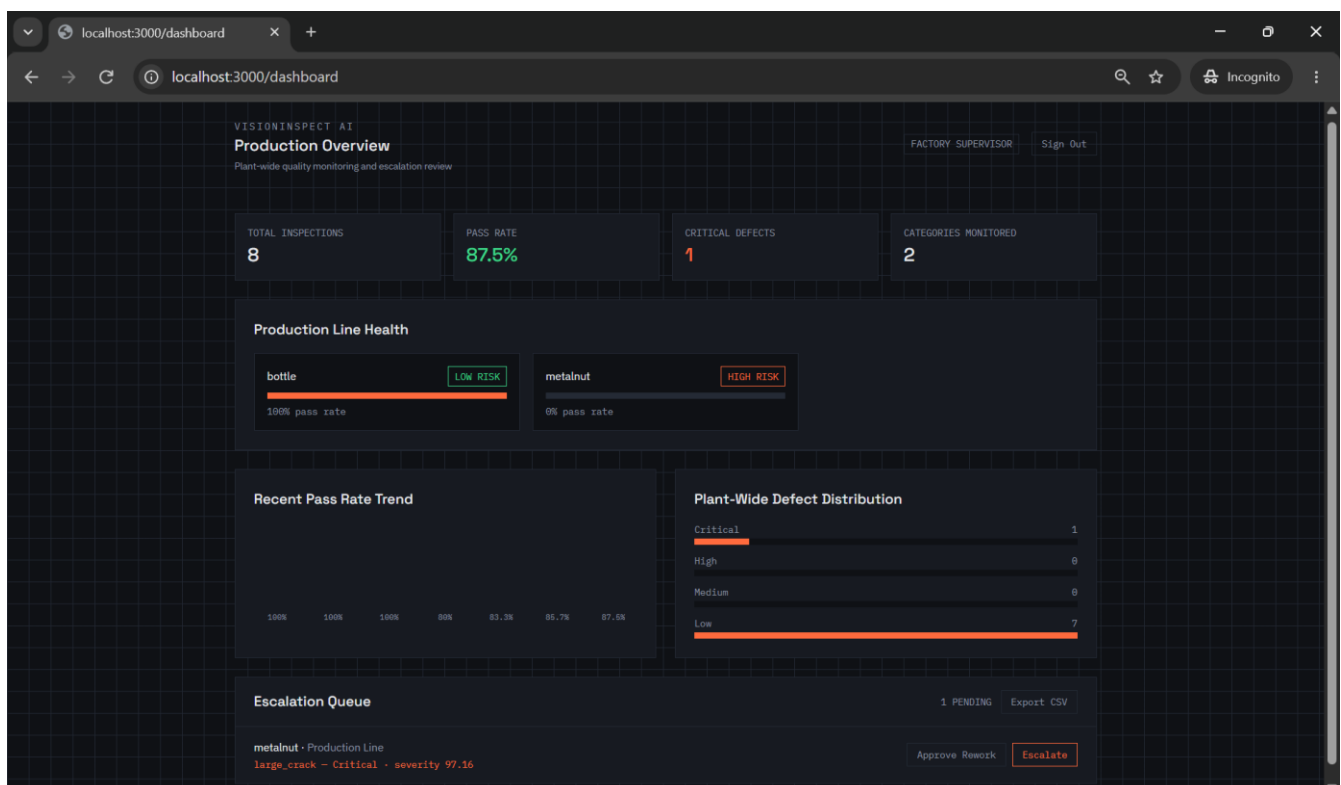
It displays:

- Total inspections
- Pass rate
- Critical defects

- Categories monitored
- Production line health
- Recent pass-rate trends
- Severity distribution
- Pending escalation queue

Dashboard data is retrieved dynamically from the backend, ensuring that displayed information reflects the current inspection database.

Critical and high-severity inspections are automatically added to the escalation queue, where supervisors can take action.



9. Supervisor Inspection Workflow

A dedicated supervisor workflow was implemented for handling critical inspections.

Supervisors can perform two actions:

Approve Rework

Marks an inspection as:

REWORK_APPROVED

Escalate

Marks an inspection as:

ESCALATED

These actions are stored in the InspectionWorkflow database table along with the inspection ID, action performed, supervisor identity and timestamp.

The workflow was successfully tested using inspection ID 5, which was approved for rework and subsequently removed from the pending escalation queue.

```
PS C:\Users\LENOVO\Documents\VisionInspectAI> Invoke-RestMethod http://127.0.0.1:8000/supervisor/workflows

inspection_id : 5
status       : REWORK_APPROVED
action_at    : 2026-08-08T08:22:49.346147
created_at   : 2026-08-08T08:22:49.348776
action_by    : factory_supervisor
id           : 1
notes        :
```

10. API Endpoints Implemented

Milestone 3 expanded the FastAPI backend with endpoints for inspection, analytics, reporting, authentication and supervisor operations.

Module	Main Endpoints
Inspection	/predict, /batch-predict, /inspect
History	/history
Analytics	/analytics, /analytics/trends, /analytics/risk-assessment
Reports	/reports, /reports/production, /report/{id}
Authentication	/auth/signup, /auth/login, /auth/profile
Supervisor	/supervisor/workflows, /approve-rework, /escalate

All endpoints are accessible through the automatically generated FastAPI Swagger interface.

11. Frontend Integration

The Supervisor Dashboard was connected to the FastAPI backend using REST API calls.

The frontend retrieves production reports, analytics, inspection history and workflow information when the dashboard loads.

Supervisor actions are also connected directly to the backend:

Approve Rework → Backend Workflow API

Escalate → Backend Workflow API

After a successful action, the dashboard updates the escalation queue automatically.

This provides a complete frontend-to-backend workflow using live database data.

12. Testing and Validation

Milestone 3 was validated through Swagger, PowerShell API requests and the integrated frontend.

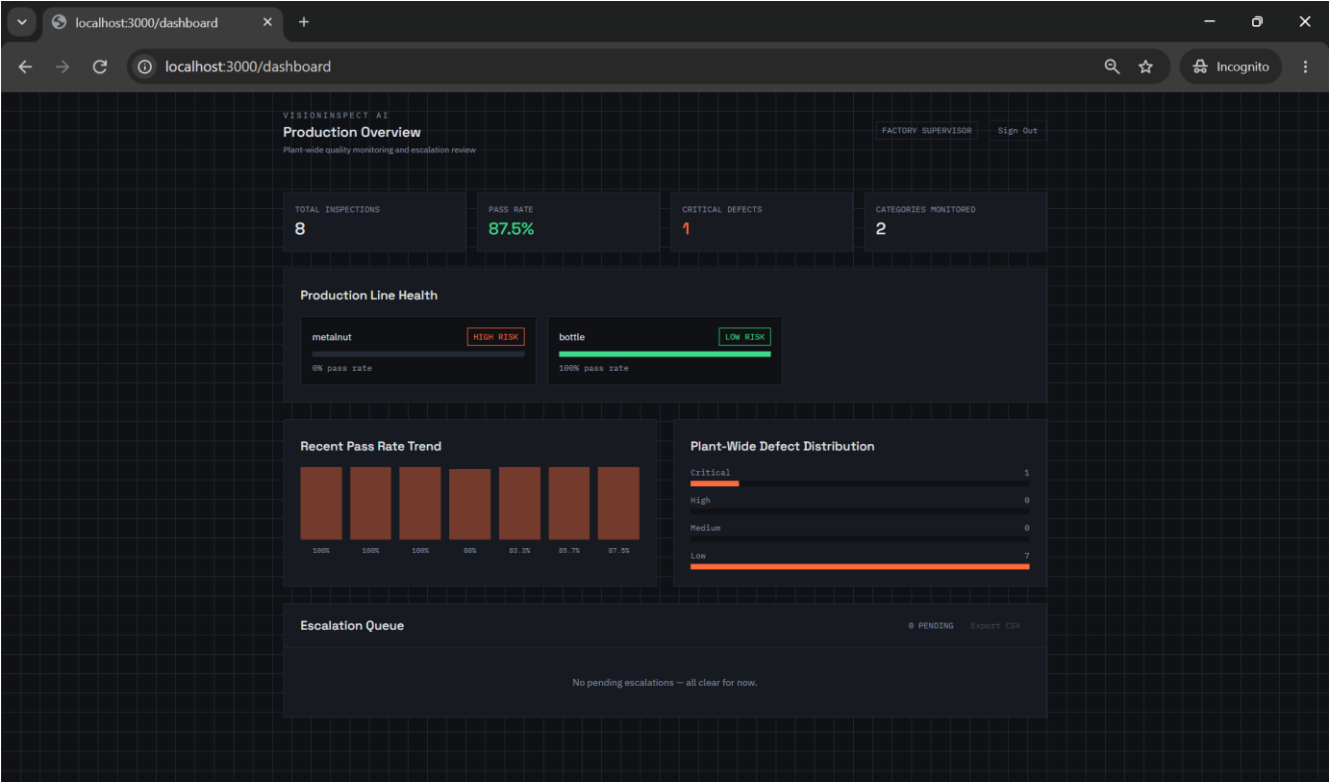
The following components were successfully tested:

Component	Result
Inspection history	Passed
Analytics	Passed
Production reports	Passed
Authentication	Passed
Supervisor dashboard	Passed
Approve Rework	Passed
Escalation	Passed
Workflow persistence	Passed

ComponentResult

Frontend API integration Passed

The /history endpoint was verified to return stored inspection records containing defect, result, confidence and severity information.



The supervisor workflow was also verified by successfully approving a critical inspection and confirming its *REWORK_APPROVED* status in the database.

13. Milestone 3 Outcomes

Milestone 3 successfully added the data management and supervisory layer to VisionInspect AI.

The major outcomes were:

- Database-backed inspection history
- Analytics and production reporting
- Risk assessment
- Authentication and user management
- Supervisor Dashboard
- Critical defect escalation
- Rework approval workflow
- Persistent supervisor actions
- Frontend-backend integration

The resulting workflow is:

Inspection → Database → Analytics → Risk Assessment → Supervisor Review → Action

14. Conclusion

Milestone 3 successfully transformed VisionInspect AI into a more complete manufacturing quality monitoring system.

The system can now store inspection results, generate production analytics, authenticate users and provide supervisors with a dashboard for monitoring and responding to critical defects.

The completed milestone provides the foundation for the remaining project work, including final AI improvements, frontend refinement and complete system integration.